Allen Mao

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Skills

AI & Simulation: Physics-Informed ML, NLP, PyTorch, VTK, PyVista
Infrastructure: AWS (EKS, ECS, EC2, SageMaker), Kubernetes, Pulumi (IaC), Nvidia Triton, Docker, CI/CD
Languages & Frameworks: Python, Java, C/C++, Go, Flask, SQL, CMake, Angular, Vue.js

Experience

Senior AI Engineer, Ansys (part of Synopsys) – Sunnyvale, CA (Remote) Jan 2024 – Present

- Architected a cloud-native data ingestion platform (VTK, Flask, Pulumi, AWS EKS) to automate mesh and metadata extraction from physics simulations, enabling scalable data prep for ML training.
- Defined "canonical data" standards to decouple fragile ML logic from raw CAD/simulation formats, implementing robust CPU-side validation to prevent expensive failures during downstream GPU training.
- Expanded platform capabilities to support transient (time-series) physics data, specifically implementing high-performance ingestion for LS-DYNA d3plot and keyfile formats using lasso-python and pydyna.
- Advanced the open-source ecosystem by upstreaming patches to lasso-python and navigating VTK C++ source and CMake builds to identify and report core library bugs to maintainers.
- Led cross-functional initiatives with Fluent, Ensight, DPF, and LSTC teams to integrate native SDKs, eliminating manual data conversion and streamlining simulation-to-ML customer workflows.

R&D Engineer II, CTO Office, Ansys – San Jose, CA (Hybrid) Sep 2022 – Dec 2023

- Refactored a proprietary, library-agnostic ML framework into a modular, object-oriented architecture, resulting in widespread adoption by internal researchers who previously relied on raw PyTorch and TensorFlow.
- Architected a multi-modal web platform (Go, Angular, Python) utilizing AWS SageMaker and Nvidia Triton to expose physics-informed ML models, enabling engineers to run ML-based physics simulation.
- Engineered a scalable research environment by deploying Jupyter Enterprise Gateway on EKS with isolated kernels, eliminating cross-library dependency conflicts and reducing environment setup time for the R&D team.

Software Development Engineering Intern, Amazon – Portland, OR (Remote) May 2021 – Aug 2021

- Modernized a legacy vendor search system by conducting stakeholder research and log analysis to architect a Vue.js and Spring application, streamlining listings management for enterprise-scale partners.

Software Engineering Intern, NASA Kennedy Space Center – Merritt Island, FL (Remote) Aug 2020 – Dec 2020

- Engineered acceptance tests for the **Artemis** Launch Control System GUI, ensuring mission-critical software reliability for the Space Launch System ground operations.

NSF REU Fellow, USC Information Sciences Institute – Marina del Rey, CA Jun 2019 – Aug 2019

- **Founding Lead** of **SoMEF** (Software Metadata Extraction Framework), an open-source NLP pipeline that has since grown to 1,100+ commits, 20 contributors, and 30+ releases on GitHub.
- Engineered the initial automated extraction logic to categorize scientific software documentation.
- **First-authored** and presented research at the **2019 IEEE International Conference on Big Data**; project continues to be utilized by the scientific community for automated model integration.

Leadership, Projects, & Honors

- **Data Tooling:** Developed a visualizer for CoNLL-formatted NLP data called view-sesame; engineered a Hackathon-winning (1st place) Chrome Extension for COVID-19 relief tracking.
- **Technical Writing & Communication:** Winner in the **Berkeley Economic Review** essay contest for analytical writing; authored a public-facing blog while working as a park ranger intern at Yellowstone National Park.

Education

University of California, Berkeley – BA in Computer Science May 2022